News Sentiment and Abnormal Return Predictability in US Stocks

Nakul Pillai

1. Hypothesis Pitch

When a company gets clearly good or bad news, the market does not fully price it in at once, so a stock's abnormal return on the first day we can act should move in the direction of the news: up after clearly positive news, down after clearly negative news.

This informs the strategy: score every headline about a stock with FinBERT, a language model trained on financial text. When a stock's average sentiment on a day is strongly positive, buy it at that day's close (or short it if strongly negative), hold it for one day, and aim to profit from the part of the reaction that has not happened yet. The news file only gives dates, not times, so the close of the article's date is the earliest point at which the trade can honestly be placed.

The mechanism is gradual information diffusion and limited attention, not a generic trend-following story. The tone of a news story is soft, qualitative information that is harder to price than a number such as an earnings surprise. Tetlock, Saar-Tsechansky and Macskassy (2008) find that negative words in firm-specific news predict returns, with prices under-reacting at first, and Chan (2003) finds that prices keep drifting after bad headlines. Attention is also limited and uneven (Barber and Odean, 2008), so not every investor reads every headline at the same moment. The evidence is not one-sided: Tetlock (2007) finds that media pessimism can reverse rather than drift. The question here is therefore narrow: after the day the news is dated, is there anything left to trade?

This project is the frozen, reproducible version of my live Sentiment Signal app, analysed in the same way as my lead-lag centrality project: parameters locked before the news data was downloaded, an in-sample period, and a final year held out and used once. The news comes from the FNSPID dataset (Dong et al., 2024). The stocks are 50 US companies drawn at random across industries rather than the lead-lag project's tech and semiconductor list, because sentiment does not depend on the stocks being connected to each other (Section 2 and Section 4.1 explain how and why the list changed).

What would prove this wrong, fixed in advance:

- Abnormal returns after a strong-sentiment day show up on the article's own day (day 0), with nothing left on day 1, the day the strategy earns. That would mean the news is already priced in by the time the trade can be placed. This is tested directly by comparing returns day by day.
- A move on day 1 that exists but disappears once realistic costs of 10 basis points round trip are charged, which matters a lot for a strategy that trades every day.
- A strategy that does not beat simply holding the same stocks.

The decision rule was written down before any results existed. The hypothesis counts as supported only if the in-sample abnormal return on the day the strategy earns rises with sentiment, positive and significant at the 5% level, **and** the out-of-sample Sharpe ratio after

costs beats the equal-weight benchmark. Anything else is reported as not supported. The strategy described here is the second design: the first was changed after its results were seen, and Section 4.1 says exactly how.

2. Data Pipeline

This study needs a long history of timestamped news for a set of stocks, matched to daily prices. The news comes from FNSPID (Dong et al., 2024), a public dataset of financial news headlines tagged by stock ticker. The prices are daily adjusted closes from Yahoo Finance for the 50 stocks, with the S&P 500 as the market benchmark.

The study window is the last 36 months of news coverage in the file. The first 24 months are the in-sample period, where the design is examined, and the final 12 months are held out and used once. Coverage was defined before looking at any results: the window ends at the last calendar month in which at least 40 of the 50 stocks have news, and a half-finished final month does not count. The file's news runs to 9 January 2024, so the window runs from 1 January 2021 to 31 December 2023. The in-sample period is 2021 and 2022, and the holdout is 2023.

A first FNSPID file (All_external.csv) was downloaded and rejected before any analysis. Its news ended in June 2020 and only 39 of the original 56 tech stocks had any news in it, so it could not meet the coverage rule. It was replaced with the Nasdaq news file (nasdaq_exteral_data.csv, about 23 GB), whose news runs to December 2023. That file is too large to work with directly, so it was read once and cut down: first to the four columns needed (date, ticker, headline, summary) for 2020 onwards, then to the stocks chosen below, and the original was deleted.

Because the first file did not cover the original tech list, and sentiment does not need the stocks to be interconnected the way the lead-lag idea does, the stock list was changed (Deviation 4 in the parameter record, written before any price or score was computed). Among the 357 stocks with news in at least 30 of the 36 window months and at least 300 articles, 80 were drawn in random order with a fixed seed, and the first 50 with usable Yahoo prices were kept. BRK and MPW had no price history under those symbols and were replaced by the next in the draw. The choice used only news counts and the seed, never prices or returns. The list is mixed: it includes large companies such as ABBV, AMGN, AMD, CRM, CSX and TMUS, and also volatile or small names such as BLNK, SPCE, CGC and BYND, and one ETF (XLF), which came out of the draw and was left in rather than hand-picked out.

The raw news needed cleaning before any sentiment was computed:

- **Ticker symbols.** Facebook's news is filed under its old symbol FB, and Alphabet's under both GOOG and GOOGL. FB was mapped to META and GOOG to GOOGL to match the price data, and any duplicates this created were dropped.
- **Duplicates.** Rows with the same ticker, timestamp and headline were dropped.
- **Roundup headlines.** Many headlines, such as "46 Stocks Moving In Friday's Mid-Day Session", are attached to every stock they list and say nothing specific about any one of them. A headline was dropped if the same title appeared on three or more

different stocks on the same day. This is counted across every ticker in the file rather than only the 50 chosen, because within a random 50 most roundups would slip through. It removed 1,042,165 of 1,748,507 articles from 2020 onwards (60%) before the stocks were drawn.

- **Summaries.** The file has a summary for each article, populated for every article in the window, so each article's sentiment is the average of its headline score and its summary score.

After cleaning, the study has 28,078 articles across 50 stocks (every stock has at least 304), made up of 24,336 unique headlines. These form 15,592 stock-days with news, 9,812 in sample and 5,780 out of sample, with a median of one article per stock-day. Figure 1 shows the sentiment distribution and the articles per month.

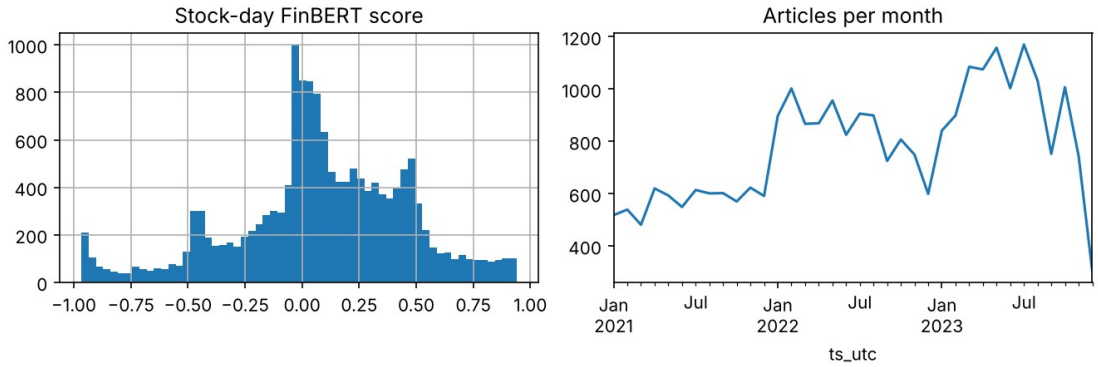


Figure 1 - Sentiment scores and news volume over the study window

Each unique headline, and each summary, is scored once by FinBERT. The score is the model's probability that the text is positive minus the probability that it is negative, so it runs from -1 to +1. VADER, a general-purpose rule-based tool, scores the same text and is kept as a comparison. A stock's sentiment on a given day is the average score of all its articles that day. FinBERT's stock-day scores average +0.08 and VADER's +0.32, so VADER leans more positive.

Each article also has to be matched to its event day, the trading day it belongs to. The plan, written before the data arrived, was that an article published before 16:00 New York time on a trading day belongs to that day, and one published later or on a weekend or holiday belongs to the next trading day. This file turned out to carry dates only: every timestamp is midnight UTC. So each article is placed on its own date, or on the next trading day if that date is a weekend or holiday, and the trade is entered at that day's close. Two consequences follow. First, the article's own day (day 0) cannot be traded, because the strategy only enters at its close. Second, because the time of day is unknown, some articles dated D were published after the close of D, and for those the entry at the close of D is slightly ahead of the information. That flatters the strategy, so any positive result has to be read with it in mind.

The universe reflects companies that still have price history on Yahoo Finance rather than a point-in-time reconstruction, so a company that was delisted or acquired during the sample is

absent by construction. That is a real limitation, and it is addressed directly in the Pitfall Checklist.

3. Strategy and Backtest

3.1 Strategy and Backtest Mechanics

A stock's raw daily return is mostly the market moving, so each return is first turned into an abnormal return: the stock's return minus its beta times the S&P 500's return. Beta is estimated over the 120 trading days up to the day before the news day, so the estimate never uses the days it is applied to. Abnormal returns are measured in basis points for the news day (day 0) and the four days after it.

Each day, the strategy averages the FinBERT scores of each stock's articles. A stock whose average is at least +0.5 is a long candidate, and one at or below -0.5 is a short candidate. This produced 2,840 signals: 1,979 in sample and 861 out of sample, 1,765 long and 1,075 short.

A signal is acted on at the close of the event day, the day the article is dated, and held for one day, so a trade earns the abnormal return of day 1. The cost is deducted on that one holding day. The trade can only use what is known by the close of day 0, and the one-day hold keeps the strategy honest about how little time there is to act on the news.

A trading cost of 10 basis points round trip is deducted from the first and only day of every trade. Capital is split equally across whichever trades are open on a given day, so the strategy never uses leverage, and days with no open trade earn nothing. A position was open on 95.6% of trading days in sample and 90.8% out of sample, so the strategy was invested almost every day, with a fresh set of trades each day.

3.2 Results

The six metrics compared across both tables are the same as in the lead-lag project, so the two studies can be read side by side:

- **Cumulative return**: the total percentage gained or lost over the period.
- **Annualized return**: the cumulative return expressed as a compound annual rate, so that periods of very different lengths can be compared on equal footing.
- **Sharpe ratio**: return earned per unit of volatility taken on.
- **Maximum drawdown**: the worst peak to trough loss along the way.
- **Sortino ratio**: like the Sharpe ratio, but it only penalises downside volatility rather than volatility in either direction.
- **Win rate**: the share of days with an open position that were actually profitable.

Two benchmarks are included alongside the strategy in both tables: an equal weighted holding of all 50 stocks in the universe, and the S&P 500 as a broad market benchmark. Both show whether the strategy actually adds anything beyond simply holding stocks.

3.2.1 In Sample Comparison

In sample, the full comparison looks like this:

Series	Cumulative Return	Annualised Return	Sharpe Ratio	Max Drawdown	Sortino Ratio	Win Rate
Strategy (with Costs)	-42.0%	-23.9%	-0.81	-48.1%	-0.89	49.9%
Strategy (No costs)	-6.2%	-3.1%	0.04	-37.2%	0.04	51.8%
Equal-weight stocks	13.6%	6.6%	0.39	-21.0%	0.61	51.1%
S&P500	2.2%	1.1%	0.15	-25.4%	0.22	49.9%

Table 1 - In-sample performance comparison.

This is also shown in the chart below:

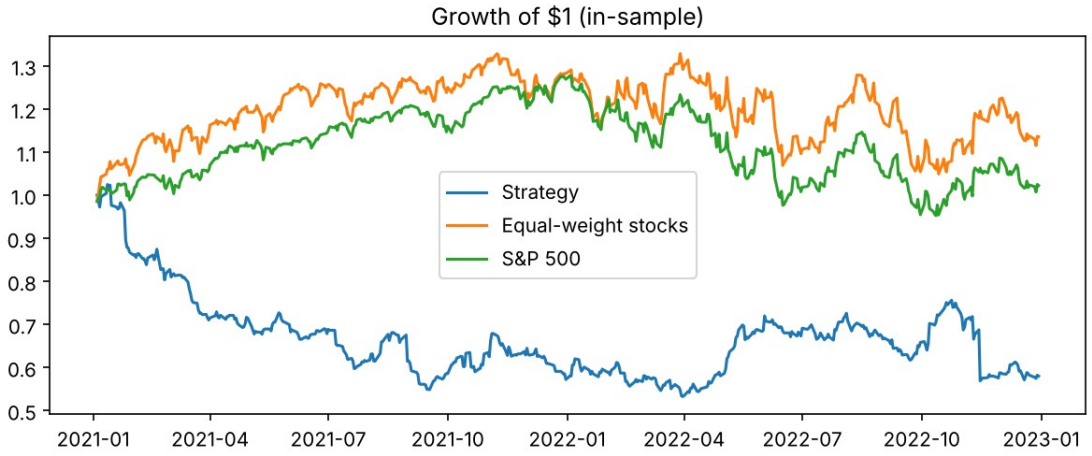


Figure 2 - Growth of $1, in-sample

In sample the strategy lost money. It fell 42.0% after costs, but only 6.2% before costs, with a Sharpe ratio of -0.81 after costs (0.04 before) and a maximum drawdown of 48.1%. The equal-weight stocks gained 13.6% and the S&P 500 gained 2.2% over the same two years. Most of the fall came in the first quarter of 2021, including a single day in late January that cost about 13%, and the worst single day was -17.3%. I did not trace which trades caused this. Late January 2021 was the retail-driven squeeze in stocks like those in the draw (BLNK, SPCE, CGC, BYND), but that link is an untested guess, not a finding.

3.2.2 Out of Sample Comparison

Out of sample, the same comparison looks like this:

Series	Cumulative Return	Annualised Return	Sharpe Ratio	Max Drawdown	Sortino Ratio	Win Rate
Strategy (with Costs)	-9.8%	-9.9%	-0.34	-20.1%	-0.51	47.1%
Strategy (No costs)	13.1%	13.2%	0.66	-12.6%	0.97	50.7%
Equal-weight stocks	16.2%	16.4%	0.94	-17.8%	1.69	55.6%
S&P500	24.2%	24.4%	1.74	-10.3%	2.85	54.8%

Table 2 - Out-of-sample performance comparison.

This is also shown in the chart below:

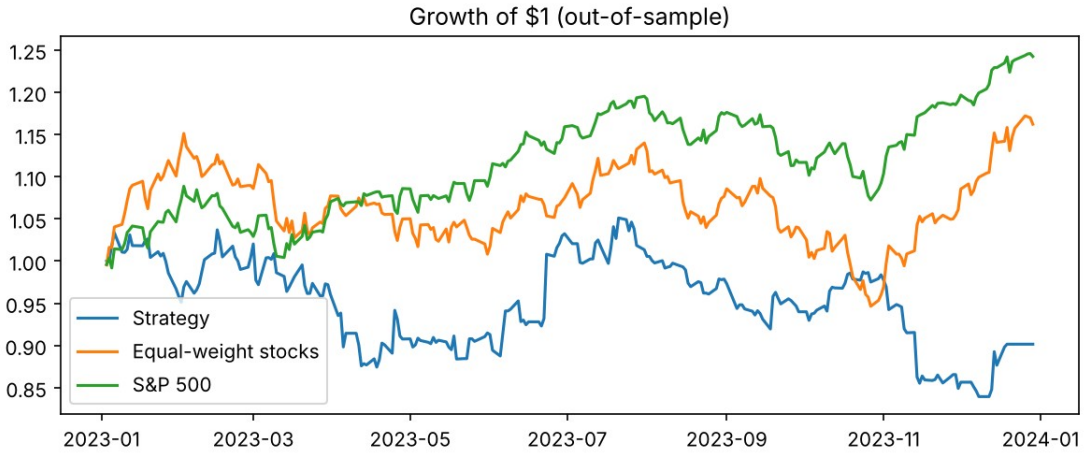


Figure 3 - Growth of $1, out-of-sample.

The in sample and out of sample cumulative returns are not directly comparable on their own, because the two periods cover very different lengths (about 2 years against 1 year held out). Annualized return puts both on the same yearly footing, and the Sharpe and Sortino ratios are already annualized, so those are the right way to compare performance across the two periods.

Out of sample the strategy lost 9.8% after costs, against gains of 16.2% for the equal-weight stocks and 24.2% for the S&P 500. Its Sharpe ratio was -0.34, against 0.94 and 1.74 for the benchmarks, and its Sortino ratio was -0.51, against 1.69 and 2.85. Its maximum drawdown of 20.1% was worse than both benchmarks (17.8% and 10.3%), and its win rate of 47.1% was below 50% and below both benchmarks (55.6% and 54.8%). In sample the picture is the same: the strategy was behind both benchmarks on every metric.

Costs are the main reason the net numbers are so poor. Before costs the strategy gained 13.1% out of sample with a Sharpe ratio of 0.66, which is respectable but still below the 0.94 of simply holding the stocks, and about flat in sample (Sharpe 0.04). Because the strategy opens new trades almost every day, a 10 basis point cost on each is a drag of roughly 36 percentage points in sample and 23 out of sample. Whatever small effect exists on day 1 is about the size of the cost, so it cannot be traded.

In one sentence: the strategy lost money after costs in both periods, and even before costs it did not beat the equal-weight benchmark in either.

4. Methodology Defence

4.1 Pre-registered Parameters

The parameters of the first design were fixed before any news data was downloaded, and none was tuned afterwards to make the result look better. The second design (Strategy B, the one this report is about) changes only the timing of the trade, as explained below. Everything else is as locked:

- beta estimated over the 120 trading days before the news day, against the S&P 500;
- a trade when a stock's average FinBERT score on a day is at least +0.5 (long) or at most -0.5 (short), from a minimum of one article;
- entry at the close of the event day, a one day hold, and a cost of 10 basis points round trip (the first design entered a day later and held three days);
- the last 36 months of coverage, with the final 12 months held out;
- the decision rule described in the Hypothesis Pitch.

That matters because the whole point of locking these choices in advance is to make the result trustworthy evidence, whichever way it comes out. A strategy that is tuned after seeing the data can be made to look good almost regardless of whether the underlying idea is real. The parameters and the decision rule were written to a dated record before the data arrived.

Five changes were made to that record afterwards. The first four were made before any sentiment score or return had been computed, and the fifth was made after the first design's results were seen. Each is logged with its reason:

1. In the first design, the primary test was moved from day 1 to days 2 to 4, because that design entered at the close of day 1 and so earned days 2, 3 and 4.
2. Ticker symbols were aliased (FB to META, GOOG to GOOGL) and a half-finished final month was excluded from the coverage rule. This was found after the first file was inspected.
3. Roundup headlines were removed, after the first file showed that a large share of its articles were multi-stock roundups.
4. The stock list changed from the 56 tech and semiconductor stocks to 50 stocks drawn at random across industries (the draw rule and seed are in Section 2), and roundup removal moved to all tickers in the file. This followed the rejection of the first news file and was done before any price for the new stocks had been downloaded.
5. After the first design (Strategy A) had been run and its results seen, the design was changed to Strategy B: the article's own date is the event day, the trade enters at that day's close and is held for one day, and the primary test is day 1. This one is not pre-registered in the usual sense, and the next paragraph explains why it was made.

The fifth change deserves a plain account. As first locked, Strategy A entered at the close of the day after the event day and held for three days. With a news file that only has dates, that meant it started trading on the third day after publication, so it tested whether a reaction keeps drifting after the news, a lag-style idea, and not the idea the project is about, which is to trade immediately on positive and negative news. Strategy A found nothing: the in-sample coefficient for days 2 to 4 was -18.2 basis points per unit of sentiment (p = 0.18), and its net Sharpe ratio was -1.35 in sample and -0.30 out of sample against 0.94 for the equal-weight stocks. Its results are kept in the project folder (outputs_A_delayed). Strategy B was then written down, with its rule and its known bias, before it was run. But it was designed after the holdout year had already been seen once, so B's out-of-sample result is exploratory, not a clean confirmation. Any positive finding would need a fresh sample to confirm, and B did not produce one.

4.2 Testing the Mechanism

The lag check in the lead-lag project asked whether the mechanism the hypothesis depends on actually exists, separately from whether the strategy makes money. The equivalent here is to regress the abnormal return on the stock's sentiment score for the news day (day 0), the next day (day 1), and the cumulative return over days 2 to 4, with standard errors clustered by stock. Day 1 is the primary test, because it is the day the strategy earns. Day 0 is a diagnostic: the strategy enters at the close of day 0, so it cannot trade it.

Period	Outcome	Coefficient (bps per unit score)	t-statistic	p-value	Observations
In sample	Day 0	110.2	4.65	<0.001	9,812
In sample	Day 1 (primary)	9.7	1.03	0.302	9,812
In sample	Days 2 to 4	-18.4	-1.44	0.149	9,812
Out of sample	Day 0	78.1	4.40	<0.001	5,780
Out of sample	Day 1 (primary)	8.1	0.96	0.339	5,780
Out of sample	Days 2 to 4	2.2	0.18	0.854	5,780

Table 3 - Abnormal return regressed on FinBERT sentiment, by day.

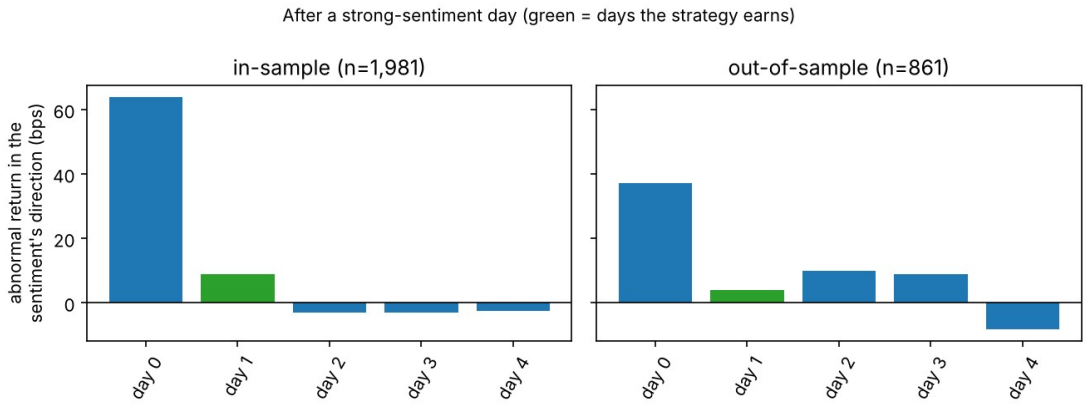


Figure 4 - Abnormal returns after a strong-sentiment day (green is the day the strategy earns).

The result is clear and not in favour of the hypothesis. On the article's own day, sentiment is very strongly related to the abnormal return: +110 basis points per unit of sentiment in sample ($t = 4.65$) and +78 out of sample ($t = 4.40$), both $p < 0.001$. By the next day, which is the day the strategy earns, that has shrunk to +9.7 basis points in sample ($p = 0.30$) and +8.1 out of sample ($p = 0.34$). The sign is the one the hypothesis predicts, but neither is significant, and neither is much larger than the 10 basis point cost. Days 2 to 4 show nothing either (-18.4 in sample, $p = 0.15$, and +2.2 out of sample, $p = 0.85$). Figure 4 tells the same story for strong-sentiment days only: the average abnormal return in the direction of the news is about +64 basis points on day 0 in sample (+37 out of sample) and only +9 (+4) on day 1. Most of the reaction has already happened by the time the trade can be placed. It is also likely that a good share of the day 0 link is the headline describing a move that had already taken place, rather than causing it, which is another reason that day cannot be traded.

Before looking at any real result, the pipeline was also checked on simulated data where the true answer is known. In one simulated world, a response of 60 basis points per unit of sentiment was planted on the day the strategy earns; the pipeline recovered 61.3 ($p < 0.001$). In a second world with nothing planted, it found 1.3 ($p = 0.59$), and the strategy's Sharpe ratio was negative. So the machinery finds an effect when there is one and does not invent one when there is not, which means a null result on the real data can be trusted as a genuine test of the idea.

4.3 Threshold Sensitivity

A threshold of 0.5 was chosen because it sits halfway along the -1 to +1 scale: strict enough that only clearly one-sided days trigger a trade, but loose enough that there are enough trades to measure anything. Rerunning the strategy at 0.3 and 0.7, and with VADER in place of FinBERT at 0.5, defends that single choice directly.

Series	Trades (in sample)	Sharpe (in sample)	Trades (out of sample)	Sharpe (out of sample)
FinBERT, threshold 0.3	4,628	-0.95	2,175	-0.17
FinBERT, threshold 0.5 (locked)	1,979	-0.81	861	-0.34
FinBERT, threshold 0.7	992	-0.59	384	-1.15
VADER, threshold 0.5	1,964	-1.13	1,131	0.13

Table 4 - Net-of-cost Sharpe ratio under alternative thresholds and models.

The verdict does not hinge on the threshold, though the picture is less one-sided than for Strategy A. After costs, no variant beats the equal-weight benchmark out of sample (0.94). The 0.3 threshold gives a net Sharpe ratio of -0.95 in sample and -0.17 out of sample, 0.7 gives -0.59 and -1.15, and VADER at 0.5 gives -1.13 and +0.13. Before costs, the looser variants look better out of sample (0.3: Sharpe 1.21; VADER: 1.05; both above 0.94), but in sample the same variants are 0.38 and -0.40, and the costs remove the gain either way. A looser threshold also means more trades (4,628 in sample against 1,979 at 0.5). The out-of-sample gross figures are the one interesting pattern here. Because Strategy B was designed after the holdout was seen, I treat them as a lead to test on new data, not a finding.

4.4 Sentiment Model Choice

FinBERT was used because general-purpose sentiment tools misread financial language. Loughran and McDonald (2011) show that most words a general dictionary labels as negative, such as "liability" or "cost", are not negative in a financial setting. FinBERT (Araci, 2019) is BERT further trained on financial text, so it has seen how this language is actually used.

The choice is tested rather than assumed. VADER (Hutto and Gilbert, 2014), a general-purpose rule-based tool built for social media text, scores the same text. On stock-days where both have a clear view they agree on the sign 71% of the time, and the two scores correlate at 0.38, so they overlap but are far from interchangeable. VADER's in-sample result was worse than FinBERT's (net Sharpe -1.13 against -0.81) but out of sample it was better (+0.13 against -0.34); neither beats the benchmark, and neither is clearly the stronger signal. The choice of sentiment model therefore does not change the conclusion.

5. Pitfall Checklist

5.1 Look-ahead Bias

Look-ahead bias is the main risk with news, because a timestamp that is wrong by a few hours can turn a prediction into hindsight. It is controlled in three places. Beta is estimated only from days before the event day. The trade is entered at the close of the event day and earns only the following day's return, so the return it earns is never in the price it was entered at. And the code checks the date rules on hand-made cases (a 17:30 Wednesday article goes to Thursday, a Saturday article to Monday when timestamps have a time, and a date-only article stays on its own date). The one place this is not fully controlled is stated openly: with dates only, an article dated D may have been published after the close of D, in which case entering at that close is slightly ahead of the information. This favours the strategy, so the poor result is not caused by it.

Take the very first in-sample trade and follow the exact pathway in order: ABBV, a long trade on articles dated 1 January 2021 (a holiday) and 4 January 2021. The beta window ends on 31 December 2020, the event day is Monday 4 January 2021, the trade is entered at the close that day, and the single holding day is 5 January. Every step only ever uses information that existed before it, in the order it would have been known in real time. This is confirmed with an explicit check in the code.

5.2 Survivorship Bias

The universe used here is a random draw from companies that still have price history today, not a reconstruction of what the market looked like at each point in the study window. Anything that was delisted, acquired or failed during the period is missing from the data, and the draw is limited to stocks that the news file covers well, so it is not a representative sample of all US stocks.

For a long-short strategy the direction of this bias is not obvious. Stocks that later failed are missing, and a short position after bad news on one of them would have been profitable, so excluding them probably understates the short side. At the same time the long side trades only survivors, which probably flatters it. The two effects pull in opposite directions, so the net bias cannot be signed with confidence. The benchmark has the same survivor-only problem, so the comparison between strategy and benchmark holds up better than either absolute figure. This is left as a known limitation. A fix would be a point-in-time constituents list that includes the companies that later disappeared.

5.3 Overfitting/ Data Snooping

The parameters of the first design, and the hypothesis and decision rule, were written down before the data was downloaded. Four changes were logged before any sentiment score or return existed, but the fifth, from Strategy A to Strategy B, was made after the first design's results (including the holdout year) were seen. It was written down and its rule fixed before B

was run, but it is the one place where the work is open to the charge of data snooping, and it is why B's out-of-sample result is exploratory.

Two further points deserve to be said plainly. The primary test is a single regression, fixed for Strategy B before B was run, but the report also shows other days, two models and three thresholds, so some of those results could look significant by chance alone. They are reported as supporting evidence, not used to decide the verdict. The threshold check in Section 4.3 is a second line of defence: no variant beats the benchmark after costs, so the conclusion does not depend on picking among them. The out-of-sample gross results for the looser variants are the only thing that looks promising, and they would be exactly the kind of result that chance, or a design chosen after seeing the data, can produce.

5.4 Transaction Costs/Slippage

A cost of 10 basis points round trip is charged on every trade, a reasonable estimate for liquid large companies. It is probably too low for the small and volatile names in the draw (such as BLNK, SPCE and CGC), so the true result would be worse. It is still a simplification: it does not grow with position size, and it ignores the cost of borrowing stock to short, which can be material for the short side of this strategy, here 1,075 of 2,840 trades.

Here costs matter a great deal. With a one-day hold the strategy opens a fresh set of trades nearly every day, so a 10 basis point cost on each trade adds up to a drag of about 36 percentage points over the two in-sample years (-6.2% before costs, -42.0% after) and about 23 over the out-of-sample year (+13.1% before, -9.8% after). The strategy trades far too often for the size of the effect it is trying to capture: the day 1 effect of about 8 to 10 basis points is no bigger than the cost of acting on it.

5.5 Out-of-Sample Validation

The final 12 months of the window were held out from the design of the first strategy and used for it once, at the end. Strategy B was designed after that, so the holdout is not clean for B. Out of sample, B returned -9.8%, with a Sharpe ratio of -0.34, against 0.94 for the equal-weight benchmark over the same period.

Out-of-sample performance was better than in sample (net Sharpe -0.34 against -0.81, and 0.66 against 0.04 before costs), but it was still negative after costs, still behind both benchmarks, and still fails the decision rule. A better out-of-sample year, on a design chosen after seeing it, is not evidence that the strategy works. 2023 was also a strong market for stocks in general, with the S&P 500 up 24%, that the strategy did not share in.

6. Efficiency Touchpoint

The slowest step in this pipeline is scoring the headlines with FinBERT, because every headline has to be passed through a transformer model. The cost grows linearly with the number of headlines scored, so a corpus ten times larger takes about ten times as long. Three things keep it manageable. Each unique headline is scored only once, however many stocks it is attached to, which matters because the same story is often filed under several tickers. Headlines are cut to 64 tokens and batched by length so the model does little wasted work on padding. And scores are saved in blocks, so an interrupted run carries on instead of starting again.

In this study, 24,336 unique headlines were scored in 29 seconds on the Mac's GPU, about 850 headlines per second, and 26,293 unique summaries in 143 seconds (184 per second), so the whole scoring step took about 3 minutes. Everything after scoring is fast because it is vectorised: matching articles to trading days and averaging by stock-day takes 0.07 seconds, computing abnormal returns for every stock-day takes 0.08 seconds, and a full backtest run takes 0.03 seconds.

At much larger scale, such as every listed stock over decades, scoring would dominate. The sensible next step would be a smaller distilled model, a GPU, or scoring only the headlines that can trigger a signal, since most stock-days with news never reach the threshold.

7. Risk Touchpoint

Risk here is measured with the Sharpe ratio and the maximum drawdown, alongside cumulative return. All three are computed separately for the in-sample and out-of-sample periods, rather than blended into one number that would hide how differently the strategy behaved in each.

The strategy did not reduce risk relative to either benchmark. In sample its annualised return was -23.9% with a maximum drawdown of 48.1%, against +6.6% and 21.0% for the equal-weight stocks and +1.1% and 25.4% for the S&P 500. Out of sample it returned -9.9% annualised with a 20.1% drawdown, against +16.4% and 17.8% for the equal-weight stocks and +24.4% and 10.3% for the S&P 500. It took more risk for a worse return in both periods.

There are two concentration risks worth naming. First, news is not spread evenly across time: on a day of market-wide news, many stocks can show strongly negative or strongly positive sentiment at once, and the strategy would open many trades that are really one bet on the market. Of the trades, 65% in sample and 56% out of sample were long, so the book leaned long, and the top five names made up 18% and 16% of trades, so no single stock dominated. Second, and this is new for a one-day hold, few trades are open at once: the median number of open trades was 4 in sample and 3 out of sample, with a maximum of 12 and 13. With capital split equally, each trade is a large share of the book, so one big move in one stock can hurt the whole strategy.

The strategy carries no leverage, but it does not hold much cash either. A position was open on 95.6% of days in sample and 90.8% out of sample, so capital was almost always at risk. The no-leverage design caps the loss on any one day, but it does not make the strategy cautious.

A plausible bad scenario brings these together: a day when one or two of the few open trades move sharply against the position. The worst single day was -17.3% in sample and -5.8% out of sample, which fits that picture, though I have not traced those days to specific trades or events.

8. Reflection

The hypothesis is not supported. Neither condition of the decision rule was met: the in-sample day 1 coefficient was +9.7 basis points per unit of sentiment ($p = 0.30$) when it needed to be significant, and the out-of-sample Sharpe ratio after costs was -0.34 against 0.94 for the equal-weight benchmark. The sign on day 1 is the one the hypothesis predicts, but the effect is small, not significant, and about the size of the trading cost. The honest scope of the result is narrow: 50 stocks drawn at random, 2021 to 2023, daily data in which the news can only be placed to the day, and a strategy that was redesigned once after its first version failed. It does not rule out a reaction within the first minutes or hours of a headline, which this file cannot show.

The most valuable finding is the size of the day 0 effect against the day 1 effect: +110 basis points per unit of sentiment on the article's own day, against +9.7 the next day. By the time the news is dated, the market has already moved, and what is left is about as big as the cost of trading it. A second finding is that costs decide the outcome for a strategy that trades every day: the out-of-sample strategy gained 13.1% before costs and lost 9.8% after. The one pattern worth testing on new data is that looser thresholds and VADER looked good out of sample before costs, but they were designed around after the holdout was seen, so they are leads, not results.

If this experiment were conducted again, I would test for the timing of the reaction directly and early, before building a full backtest around it. A few further extensions would make this work more robust:

- **Intraday timing**: this news file only has dates, so any reaction within minutes or hours of a headline is invisible, and articles dated D could have come after the close of D. A source with real timestamps, and intraday prices, would test "trade immediately on the news" properly, and it is the most important extension.
- **Survivorship bias**: use a point-in-time constituents list, so stocks that were delisted or acquired are included rather than excluded by construction.
- **Broader parameter sensitivity**: only the sentiment threshold and the model were stress tested for Strategy B. The beta window and the minimum number of articles were never varied, and a hold longer than one day was only tried in the first design.
- **A fresh sample for any lead**: the out-of-sample gross results for looser thresholds and for VADER would need to be tested on a later period or on different stocks before anyone believes them.
- **Realistic slippage and shorting costs**: a size and liquidity dependent cost model, with borrowing costs on the short side, would be a more realistic assumption than a flat 10 basis points.
- **Story novelty and source quality**: a headline repeating a story already published an hour earlier carries no new information. Weighting by source, or scoring only the first headline about an event, might sharpen the signal.

- **A longer, more recent sample**: the window ends where the news file ends, so a newer news source would test whether the result holds in more recent markets.

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